中南大学 2003-2004 学年第二学期《生物化学》考试试卷

Exam paper for the 2nd term of 2003-2004 at Central South University

Subject: Biochemistry
Specialty：Bioengineering
Course character：Required
Exam mode：Without reference documents but permission of any kind of dictionary
Exam time length：120 minutes

Student Name_____, Register No. _____ Score_____

Part I True or False (1 point each， 20 pts)

Please write T (true) or F (false) in the blanket

1. Passive diffusion needs energy ( )
2. Glucose transport in erythrocytes occurs by facilitated diffusion ( )
3. All active transport systems are energy-coupling devices ( )
4. Glycolysis is a catabolism to degrade glucose ( )
5. Glycolysis provide the organism not only energy but also some important intermediates ( )
6. TCA cycle is the major way to produce energy by aerobic organisms ( )
7. Both photophosphorylation and oxdative phosphorylation produce ATP ( )ATP is not the major energy form in the living organism ( )
9. ATP, acetyl-CoA and glucose-6-phosphate have high-energy bond. ( )
10. NADH normally occurs in biosynthesis pathway, while NADPH occurs primarily in biodegradation pathway ( )
11. TCA cycle produces ATP directly ( )
12. Carbohydrate produced by photosynthesis occurs in the dark period ( )
13. Lipid bilayer takes part in mass transfer, energy production and signal communication ( )
14. Biochemical reaction has not space but substrate specifications ( )
15. Degradation of lipid can provide much more energy and water than carbohydrates ( )
16. NADPH is necessary for maintaining the reductive environments for blood cells ( )
17. Proline often occurs in a terminal and turn of α-helix of protein ( )
18. Alpha helix and beta-sheet of protein is due to hydrogen bond, but subunits of hemoglobulin are arranged due to hydrophobic force. ( )
19. Change in conformation is normally accompanied by breakage of chemical bond(s) ( )
20. Pre-mRNA transcripted from DNA for prokaryotes normally does not need any processing to form the functional mRNA. ( )

Part II Multiple Choice (1 point each， 30 pts)

Choose the one best answer to each of the following questions.

1. ____ Biochemical synthesis of new cell material is called

A. Metabolism
B. Anabolism
C. Catabolism
D. Synthatabolism

2. ____ TCA cycle is a pathway of

A. anabolism
B. catabolism
C. photosynthesis
D. reductive pathway

3. ____ Chemoorganotrophs use which of the following as an energy source?

A. Organic compounds
B. Inorganic compounds
C. Both organic and inorganic compounds
D. Light

4. ____ A fastidious organism is

A. oxygen sensitive
B. temperature sensitive
C. nutritionally demanding
D. atmospheric pressure sensitive

5. ____ If G_0 is positive

A. the reaction is exergonic and requires the input of energy.
B. the reaction in endergonic and requires the input of energy.
C. the reaction is exergonic and energy will be released.
D. the reaction is endergonic and energy will be released.

6. ____ Which statement is true?

A. Coenzymes are generally bound tightly to their respective enzymes.
B. Prosthetic groups are generally bound tightly to their respective enzymes.
C. Coenzymes and prosthetic groups are both bound tightly to their respective enzymes.
D. Coenzymes and prosthetic groups are technically not bound to any enzymes.

7. ____ In an electron carrier system, the net energy change is determined by the difference in reduction potentials between the

A. primary electron donor and the terminal electron donor.
B. primary electron acceptor and the terminal electron acceptor.
C. primary electron acceptor and the terminal electron donor.
D. primary electron donor and the terminal electron acceptor.

8. ____ The Embden-Meyerhof Pathway is another name for

A. the citric acid cycle
B. glycolysis
C. electron transport

D. NADH production

9. ____ In glycolysis

A. the crucial product is ATP; the fermentation products are waste products.

B. the crucial product is ethanol or lactate; ATP is a waste product.

C. the crucial product is CO2; ATP is a waste product.

D. the crucial product is not relevant because glycolysis is not a major pathway.

10. ____ In chemiosmosis

A. OH- accumulates on the outside of the membrane while H^+ accumulates on the inside.

B. OH- accumulates on the inside of the membrane while H^+ accumulates on the outside.

C. both OH- and H^+ accumulate on the inside of the membrane.

D. both OH- and H^+ accumulate on the outside of the membrane.

11. ____Which statement is not true? Uncouplers prevent ATP synthesis

A. by interfering with electron flow.

B. by creating leaky membranes.

C. without affecting electron transport.

D. by destroying the proton motive force.

12. ____ The chemical substance that enters the citric acid cycle for further metabolism is

A. ethyl alcohol

B. acetyl-CoA

C. adenosine triphosphate

D. pyruvic acid

13. ____ Which of the following occurs in the absence of oxygen?

A. Fermentation

B. Anaerobic respiration

C. Anoxygenic photosynthesis

D. All of the above.

14. ____ Most chemolithotrophs are

A. phototrophs

B. autotrophs

C. heterotrophs

D. organotrophs

15. ____For a carbon source, chemoorganotrophs generally use such compounds as

A. glucose

B. carbon dioxide

C. triglycerides

D. sunlight

16. ____ For each molecule of pyruvate that enters the citric acid cycle, how many molecules of CO2 are generated?

A. 1

B. 2

C. 3

D. 6

17. ____ Whether an organism is classified as a photoheterotroph or a photoautoroph

depends on its

A. energy source.

B. carbon source.

C. oxygen requirements.

D. All of the above.

18. ____ How many ATP result from the transfer of electrons from NADH to O_2 through the electron transport chain?

A. 1

B. 2

C. 3

D. 6

19. ____To describe a type of photosynthesis as "oxygenic" implies that

A. oxygen is produced

B. oxygen is consumed

C. oxygen functions as a catalyst

20. ____ The Calvin cycle

A. is responsible for the fixation of CO_2.

B. utilizes only ATP.

C. requires phosphofructokinase and pyruvate decarboxylase

D. all of the above

21. ____ DNA is always synthesized in the ____ direction.

A. amino terminal to carboxy terminal

B. carboxy terminal to amino terminal

C. 3' to 5'

D. 5' - 5'

E. 5' - 3'

22. ____ The short pieces of DNA made in the discontinuous strand are called _____ fragments?

A. Intron

B. Cistron

C. Exon

D. Polycistronic

E. none of these

23. ____ DNA replication begins at locations called ___ sites.

A. Promoter

B. Ori

C. Shine-Delgaro

D. Okazaki

E. Intron

24. ____ All DNA polymerases need a _____ to attach a nucleotide to.

A. RNA primer

B. Ligase

C. 3'-OH

D. deoxynucleotide

E. 5'-triphosphate

25. ____ Which of the following is not a substrate for DNA polymerases?

A. deoxyadenosine triphosphate

B. DNA template

C. a nucleotide primer with a free 3'-OH

D. deoxyuracil triphosphate

E. deoxyguanosine triphosphate

26. ____ Which of the following replaces the RNA primer with deoxynucleotides?

A. DNA polymerase I

B. DNA polymerase II

C. DNA polymerase III

D. DNA ligase

E. primase

27. ____ If Vmax=100μmol/mL sec and Km=2mM, the velocity of the reaction is _____, when [S]=20mM

A. 108 μmol/mL sec

B. 91 μmol/mL sec

C. 67 μmol/mL sec

D. 145 μmol/mL sec

28. ____ The carboxylation catalyzed by a carboxylase usually needs a prothetic group

A. Fe^{2+}

B. biotin

C. ferrocyanin

D. Thiamine

29. ____ In a typical cycle of beta-degradation of C16 saturated fatty acids, it includes two steps of oxidation, in which the electron carriers are respectively :

A. FAD and NAD^+

B. NAD^+ and FAD

C. FAD and FAD

D. NAD^+ and NAD^+

30. ____In an erythrocyte undergoing glycolysis, what would be the effect of a sudden increase in the concentration of ATP

A. Glycolysis inhibited

B. glycolysis stimulated

C. fructose-1,6-bisphosphate accumulated

D. acetyl-CoA accumulated

Part III Filling Blankets (1 point each, 20 pts)

Please fill each blanket with suitable word(s) either in English or Chinese

1. When external electron acceptors are not available in anoxic environments, carbon is catabolized by (_____).
2. The two major products of pentose phosphate pathway are (_____) and

(_____)

3. The major chemical energy forms from catabolism that are used for biosynthesis are (_____) and (_____).

4. There are six subclasses of enzymes according to the Enzyme Commission, please give their systematic names (_____; _____, _____, _____; _____; _____)

5. The fates of pyruvate produced by glycolysis are :
(1)_____; (2)_____
(3)_____; (4)_____

6. The energy source for the heterotrophs, phototrophs and lithotrophs correspondingly are (_____; _____; _____).

7. Besides glycolysis, please list the other two most important pathways:
(_____; _____)

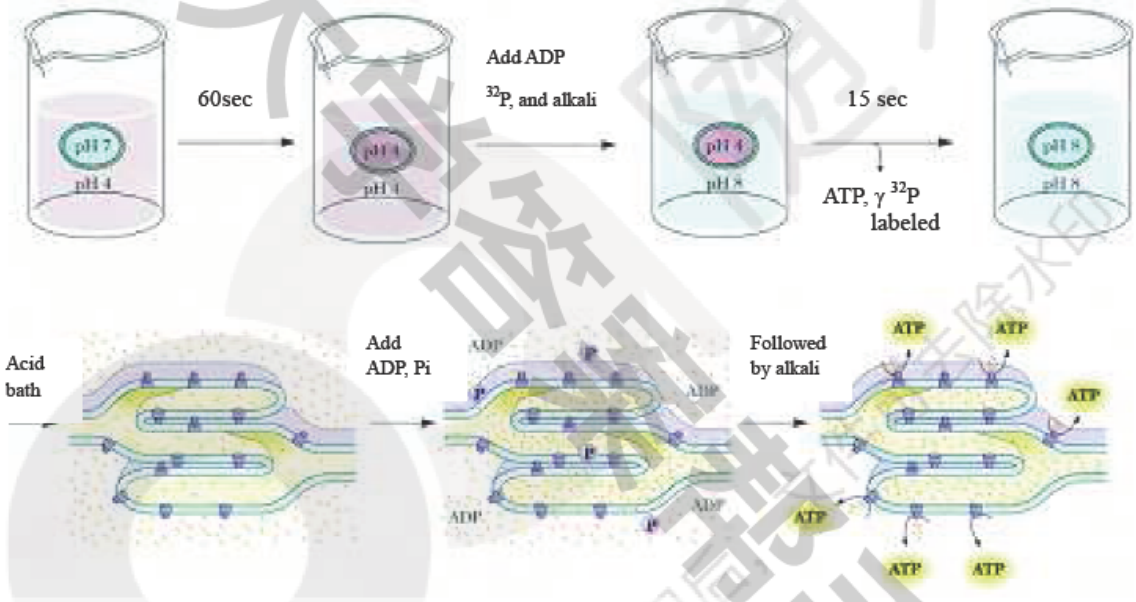
Part IV Short essay questions (30 pts)

Please answer the following short essay questions either in English or Chinese

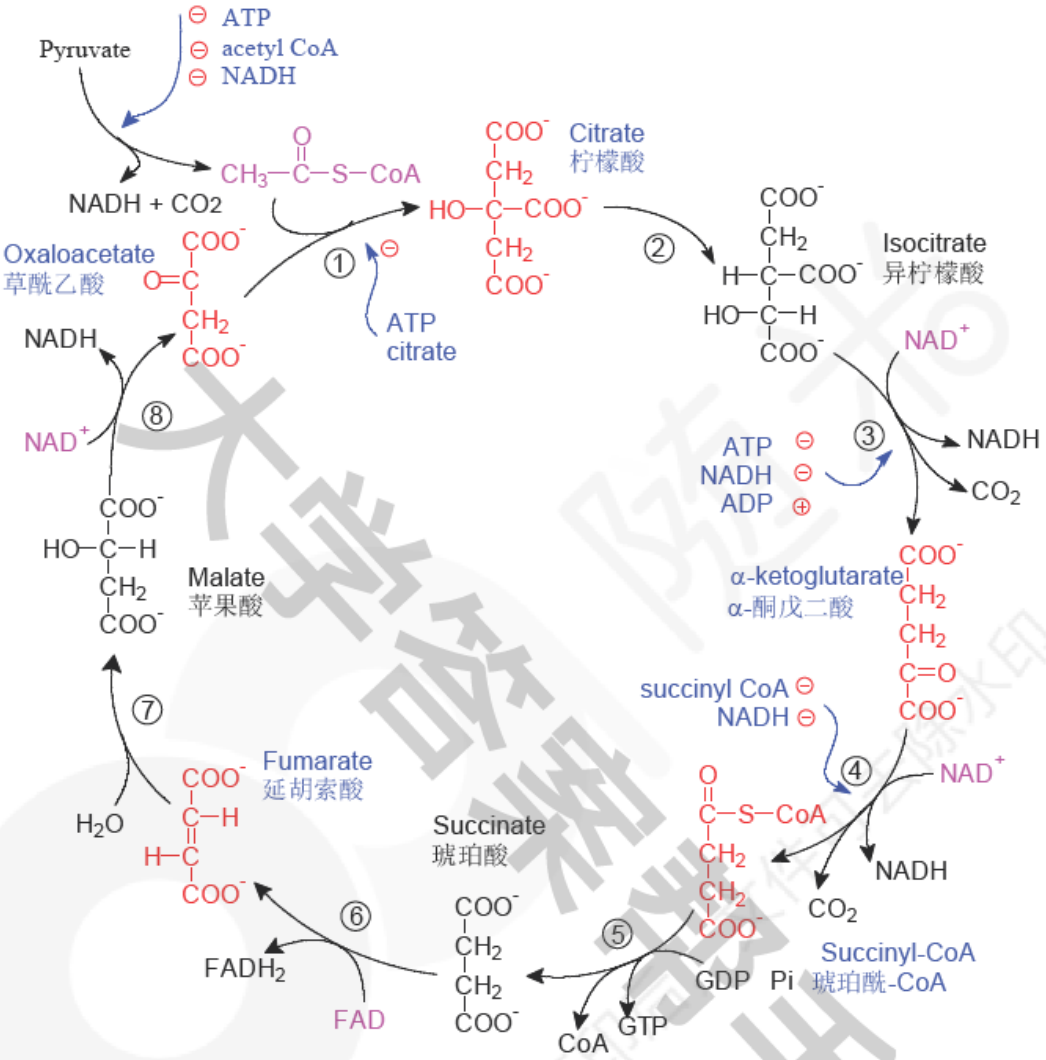
1. Please describe the structural characters of the four different levels of protein.(4 pts)

2. **Please compare and contrast fermentation and respiration. (4 pts)**

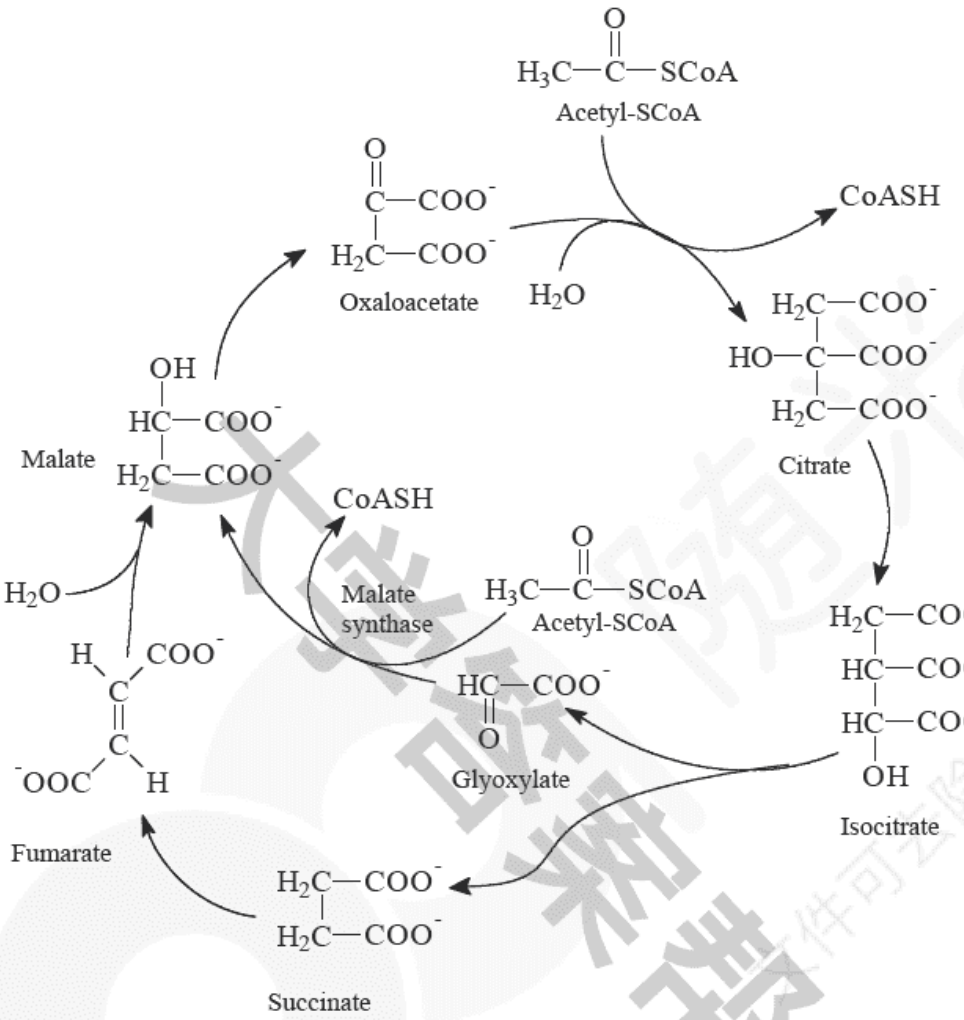
3. **Chemiosmotic is used to explain the mechanism of phosphorylation (see the following figure). Please explain. (4 pts)**



4. Please describe the regulation of the TCA cycle according to the following figure (4 pts)

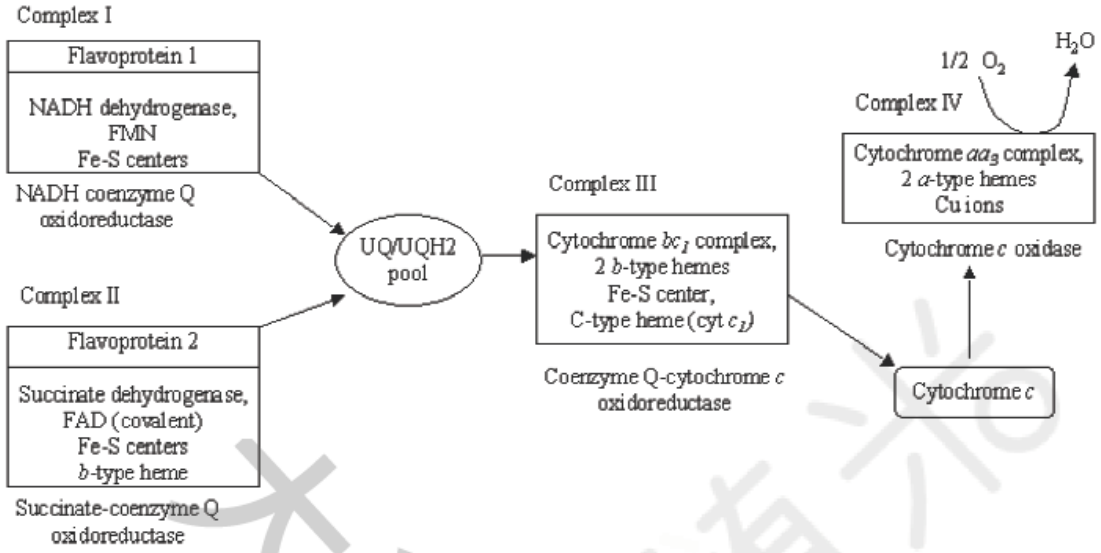


5. The glyoxylate cycle of plants and bacteria (see figure below) are very important for their seedling, Please explain (4 pts).



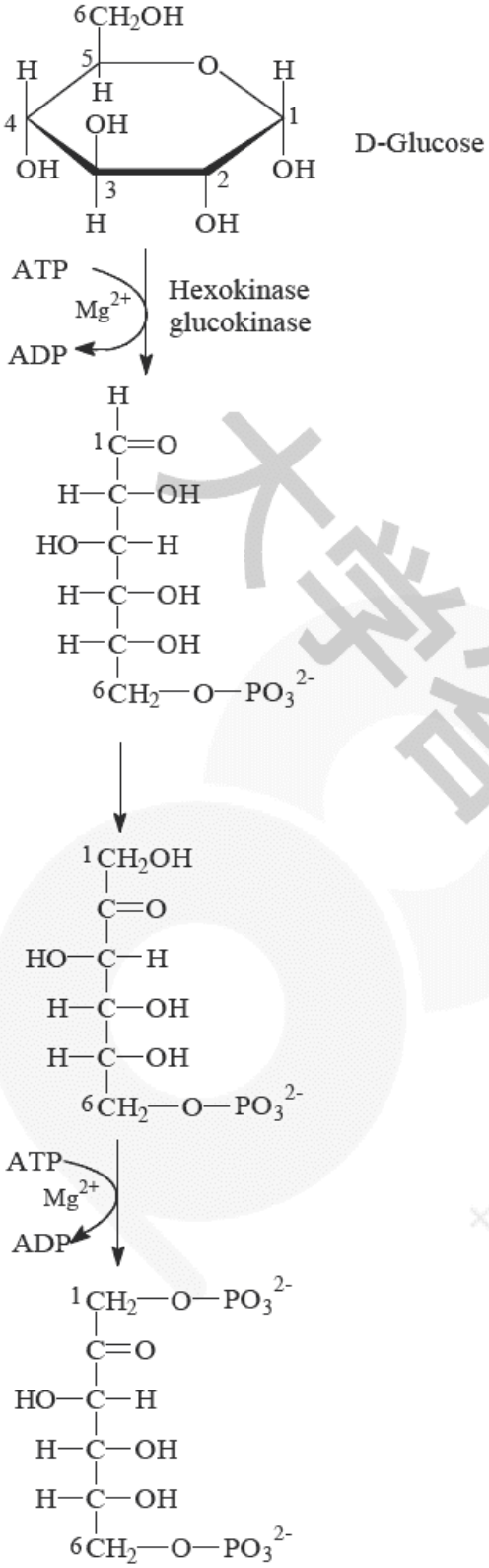
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6. Describe the complexes and pathways in the mitochondrial electron transport chain according to the following figure. (5 pts)



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7. Please give the enzyme names for the 2nd and 3rd steps, and also give the names of products for all steps listed. (5 pts)



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参考答案

Subject: Biochemistry　　Specialty：Bioengineering

Part I　True or False　(1 point each，20 pts)

1F　2T　3T　4T　5T　6T　7T　8F　9F　10F　11F　12T　13T　14F　15T　16T　17T　18T　19F　20T

Part II　Multiple Choice　(1 point each，30 pts)

1B　2B　3A　4C　5B　6B　7D　8B　9A　10B　11A　12B　13D　14B　15C　17B　18C　19A　20A　21E　22E　23B　24A　25D　26A　27B　28B　29A　30A

Part III　Filling Blankets (1 point each,　20 pts)

1. (_fermentation_).
2. (_NADPH___)　(___ribose-5-phosphate, or ribulose-5-phosphate___)
3. (___ATP__)　(___NADPH____).
4. (__oxidoreductase___; _____transferase_____,
   _____hydrolase_____, _____lyase_____;
   _____isomerase_____; _____ligase_____)
   (1)____to TCA cycle_____; (2)_____to form ketone bodies_____
   (3)_____to form ethanol and CO_2_____; (4)_____to form lactate_____
   (_____organic compounds_____; _____inorganic compounds_____;
   _____light source_____).
6. (_____TCA cycle_____; _____pentose phosphate pathway_____)

Part IV　Short essay questions (30 pts) (略)